SOFTWARE QUALITY ENGINEERING

Lab 7: Pairwise Test Design

Book Search Form — 2-way Covering Array

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Project	LMS Book Search Form
Tool	allpairspy 2.5.1

1. Objective

Pairwise testing selects a smaller set of test cases so every possible pair of values across every two parameters appears at least once. It reduces the number of tests compared with trying every full combination.

2. Parameters and Values

Parameter	Value 1	Value 2	Value 3
P1 — Query type	Title	Author	ISBN
P2 — Filter	None	Available	Genre
P3 — Sort order	Relevance	A-Z	Date
P4 — Page size	10	20	50

3. Generated Pairwise Test Cases

TC	P1: Query	P2: Filter	P3: Sort	P4: Size
PW1	Title	None	Relevance	10
PW2	Author	Available	A-Z	10
PW3	ISBN	Genre	Date	10
PW4	ISBN	Available	Relevance	20
PW5	Author	None	Date	20
PW6	Title	Genre	A-Z	20
PW7	Title	Available	Date	50
PW8	Author	Genre	Relevance	50
PW9	ISBN	None	A-Z	50

Generated output: 9 test cases; 54 required pairs; 0 missing pairs; 81 full combinations; 9x reduction.

4. P1 and P4 Verification

The table below confirms that all nine combinations of query type and page size appear in the generated cases.

P1 Query type	P4 = 10	P4 = 20	P4 = 50
Title	PW1	PW6	PW7

P1 Query type	P4 = 10	P4 = 20	P4 = 50
Author	PW2	PW5	PW8
ISBN	PW3	PW4	PW9

5. Removal Experiment

The final test case (PW9: ISBN, None, A-Z, 50) was temporarily removed. The script then reported 8 test cases and 6 missing pairs. Two missing pairs are:

Parameter pair	Missing value pair	Reason
P1 and P2	ISBN + None	In the provided generated array, this pair appears only in PW9.
P1 and P4	ISBN + 50	In the provided generated array, this pair appears only in PW9.

After deleting the temporary cases.pop() line, the script was run again and returned 9 cases, 54 required pairs, and 0 missing pairs.

6. Reduction Calculation

Full combinations = 3 × 3 × 3 × 3 = 81

Pairwise test cases = 9

Reduction factor = 81 ÷ 9 = 9. The pairwise suite uses one-ninth as many test cases as exhaustive testing, a reduction of 72 test cases (about 88.9%).

7. Screenshot to Insert



